The galactic acceleration scale as the Jeans rate of the vacuum

Mike One · research note · 24 October 2026, updated 2 November 2026

The finding

The galactic acceleration scale (a_0, called $g_\dagger$ here) has been known for about 40 years to lie close to $cH_0/2\pi$, with no accepted reason why. This note proposes one and tests it.

Proposal. The vacuum is a physical medium. Any self-gravitating medium of density ρ has a natural gravitational rate, the Jeans frequency $\omega_J = \sqrt{4\pi G\rho}$. Applied to the vacuum's energy density ρ_Λ:

$$\omega_J = \sqrt{4\pi G\rho_\Lambda} = c\sqrt{\Lambda/2}, \qquad g_\dagger = \frac{c\,\omega_J}{2\pi} = \frac{c\sqrt{4\pi G\rho_\Lambda}}{2\pi} = 1.056\times10^{-10}\ \mathrm{m\,s^{-2}}.$$

The commonly quoted value is $1.20\pm0.02\,(\mathrm{random})\pm0.24\,(\mathrm{syst})\times10^{-10}\ \mathrm{m\,s^{-2}}$ (McGaugh et al. 2016); 1.056 lies inside it, and SPARC with stellar masses free per galaxy gives 1.054.

Mechanism. Matter relaxes the vacuum wherever matter's own gravitational rate, g/c, exceeds the vacuum's Jeans rate. The transition sits where the two rates are equal.

Why it looks like cH_0 today. ρ_Λ dominates the present expansion, so the vacuum's clock and the expansion clock agree today to 1.4%. In the past H was larger but ρ_Λ was not, so this scale stays constant while $cH(z)/2\pi$ grows.

The derivation (one postulate)

Postulate. The relaxed vacuum is the vacuum in its ground state; the unrelaxed vacuum is thermally excited by the vacuum's own bath. Matter's pull sets a vacuum mode; the vacuum sets the bath.

1. **Bath.** The vacuum medium has a Jeans rate ω_J, and its horizon-type temperature is $kT = \hbar\omega_J/2\pi$, the form of the Gibbons–Hawking temperature. The 2π belongs to the bath.
2. **Statistics.** For a bosonic mode in a thermal bath, $P(\text{ground state}) = 1-e^{-x}$. The relaxed fraction is $r = 1-e^{-x}$, and $\nu = 1/r$ gives $\nu - 1 = \langle n\rangle = 1/(e^x-1)$.
3. **Symmetry.** Far from matter the unrelaxed vacuum is invariant under shifts along the scale coordinate, i.e. under $(t,r)\to\lambda(t,r)$. With $x = (g_N/g_\dagger)^p$ the deep regime gives $g \approx g_N^{1-p}g_\dagger^{p}$; scale invariance forces $p=\frac{1}{2}$: the mode energy is the geometric mean of matter's rate and the bath frequency.
4. **Result,** with no adjustable factor: $\nu = 1/\left(1-e^{-\sqrt{g_N/g_\dagger}}\right)$, the observed radial acceleration relation (McGaugh et al. 2016), with its scale.

Supporting structure (derived along the way)

1. The galaxy law $\nu(y) = 1/(1-e^{-\sqrt{y}})$ is a Bose–Einstein occupation, $\nu-1 = 1/(e^x-1)$ with $x = \sqrt{g_N/g_\dagger}$; its field action is the Newtonian term plus $4\int_0^{\sqrt{y}} u^3/(e^u-1)\,du$, a Planck (blackbody) integral cut at $\sqrt{y}$.
2. At the midpoint on the scale coordinate between any mass's black-hole radius and the vacuum horizon, the Newtonian pull is $cH_\Lambda/2$, independent of the mass: the reason a single universal galaxy scale exists.
3. $\sqrt{\Lambda/2}$ is the expansion rate at the moment the vacuum is twice the matter, the moment of the complete sign flip, $q=-\frac{1}{2}$, the mirror of the matter era's $q=+\frac{1}{2}$.

Relation to prior work

Milgrom (1999) ties MOND inertia to the Unruh temperature of an accelerated observer in de Sitter space, using temperature differences; he obtains $a_0 \sim (\Lambda/3)^{1/2}$ forms, interpolating functions that differ from the observed one, and no mechanism fixing the scale. This work uses occupation statistics (relaxed = ground state), derives the observed function, and has the Jeans factor $\sqrt{\Lambda/2}$; the $\sqrt{\Lambda/3}$ form fails three of the datasets below.

Hyperconical modified gravity (HMG; Monjo 2018–2025) derives $a = 2c/(\gamma_0 t)$ from the projection of a hyperconical universe, with $\gamma_0 \approx 13 \pm 3$ required by galaxies. With HMG's $t = 1/H$ and $\Omega_\Lambda = 0.6937$, the Jeans scale is exactly this form with $\gamma_0 = 4\pi/\sqrt{3\Omega_\Lambda/2} = 12.3$, and the implied projection angle, 0.4624π, falls between HMG's two fitted values. The two differ in time: HMG's scale goes as $1/t$, the Jeans scale is constant. This link is developed in the paper *Hyperconical gravity applied to the renormalization scale* (github.com/THEREALMIKEONE/Hyperconical-gravity-renormalization-scale-Repository).

Tests

All rules were fixed before running; no parameter was fitted to the data tested.

Dataset	Jeans (constant)	Expansion $cH(z)/2\pi$	$\sqrt{\Lambda/3}$ form
SPARC, 153 rotation curves (ΔBIC vs free fit)	-7.8	-7.2	$+53$, fails
Mistele et al. 2024 lensing velocities ($\Delta\chi^2$)	$+3.2$, passes	$+0.9$, passes	$+19.0$, fails
KiDS-1000 late-type and blue lenses ($\Delta\chi^2$)	$+1.2$, passes	$+0.2$, passes	$+13.8$, fails
High-z kinematics, 145 galaxies, $z = 0.7$–2.5	consistent (1.7σ)	excluded (3.4σ)	consistent

The Jeans formula is the only candidate consistent with all four datasets.

Search history (for honesty)

About 100 candidate formulas were tested on SPARC first; none passed galaxies, redshift and a physical mechanism together. Ruled out: the plain vacuum horizon temperature, the expansion rate, the deceleration rate and parameter, the Hayward–Kodama temperature, the conformal coupling $\xi = 1/6$, the standard de Sitter factors ($\sqrt{2}$, $3/2$, 2) and several Bose–Einstein counting factors. Tried as derivations of the factor 2 and closed: the brane junction equation, Milgrom's temperature mechanism, and a scalar-field solution along the scale coordinate. The Jeans reading came after this search and matches the best SPARC candidate, so part of its SPARC score reflects selection; the two lensing tests were run afterwards with rules fixed first.

Open points

- The chain rests on one postulate (relaxed vacuum = ground state), testable only by its predictions.
- At $z < 0.5$ the Jeans and expansion forms differ by about 0.05 dex; their separation rests on the high-z test, whose four samples disagree with each other ($\chi^2 = 9.9$ for 3 degrees of freedom).
- The two lensing datasets share KiDS lenses, so they are not fully independent.

Decisive future tests

1. Late-type weak lensing at $z \gtrsim 1$ (Euclid, Rubin): the Jeans scale is constant; the expansion form is about 1.8 times larger at $z = 1$.

2. The scale follows ρ_Λ: with $\Omega_m h^2$ fixed by the CMB, $H_0 = 73$ instead of 67.4 km s^{-1} Mpc^{-1} would raise it by about 12%, against 8% for $cH_0/2\pi$.
3. Stellar mass-to-light ratios to about 0.05 dex would separate 1.056 from nearby candidates.

Files

The scripts in this repository run every test above on public data (SPARC, and R. Monjo's lensing_RAR_KiDS compilation); see the README.